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Dr. Virendra Swarup Institute of Computer Studies

(An Institute of Computer Application & Management)

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Synopsis on

ONLINE ATTENDANCE MANAGEMENT SYSTEM

FOR

Student Attendance Management

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Attendance Management System

Project Title

The title “Attendance Management System” is illustrated from the conventional method of marking the attendance of the students.

Student Attendance Management System is an online platform designed to facilitate the efficient recording and management of student attendance by faculty members. Additionally, the system enables students to verify the accuracy of their attendance records, ensuring transparency and accountability.

The system ensures complete confidentiality of all users by safeguarding any personal information they provide, maintaining strict privacy and data protection standards.

Introduction of the Project

"Attendance Management System" is illustrated from the conventional method of marking the attendance of the students.

Student Attendance Management is an online attendance management portal that provides the faculties with the facilities to mark the attendance of the students. In addition to this, the students can also check whether they are correctly marked present.

We also guarantee to maintain 100% confidentiality to all the users of this application about the personal information shared by them.

Brief description of the project

The Attendance Management System portal is a use to mark and check the attendance of the students. In addition to this, the students can also check whether they are correctly marked present. It is a web-based application used to maintain attendance of students.

As soon as a teacher or a student joins the institution, he or she is requested to sign up to the online attendance management system. After this, admin is required to review and approve their account activation. As their account gets approved, they can access their account.

The application can be accessed by the teachers, coordinators, heads of the department and dean who fall under the category of college faculty and by the students and their parents. Super user admin is the root node who adds the students into the system and accesses every part of the system.

The project provides easy access to modules, such as viewing the faculties' profile, students' profile, view attendance of students and number of students present on a particular day. The User interface is very simple to deal with. Data storing and recovery is fast and secure. Additionally, graphic reports are available which make interpretation of data simple and easy to grasp. Various reports such as Student-wise attendance, Day-wise attendance, Class-wise attendance, Month-Wise, Class-Attendance, etc. can be easily generated. Current and back-dated reports are available readily. The application claims to provide reduced operational time, increased accuracy and reliability, increased operational efficiency and data security. There are several attributes of software that can serve as requirements. In case of student attendance management these are reliability, availability, security, maintainability and portability.

The project will have a very vast scope in future. The project can be implemented on intranet in future. Project can be updated in near future when requirement for the same arises, as it is very flexible in terms of expansion. With the proposed software for database space managers ready and fully functional the client is nowable to manage and hence run the entire work in a much better, accurate and error free manner.

Objectives

- ✓ To build a web-based application to maintain attendance of students.
- ✓ Reduce the paperwork and elimination of errors due to automated attendance.
- ✓ Reduce operational time.
- ✓ Increase accuracy and reliability
- ✓ Increase operational efficiency
- ✓ Maintain data security
- ✓ Provides access to:
    1. Students
    2. Teachers
    3. Dean
- ✓ To access the application by providing the valid ID and Password of the registered users of the application.
- ✓ Easy access to modules, such as viewing the faculties' profile, students' profile, view attendance of students and number of students present on a particular day.
- ✓ To make the application user friendly
- ✓ Provide fast and secure data storing and recovery

- ✓ Availability of graphic reports which makes interpretation of data simple and easy to grasp

- ✓ Easy generation of reports such as Student-wise attendance, Day-wise attendance, Class-wise attendance, Month-Wise, Class-Attendance, etc.

Current and back-dated reports are available readily

Project Category

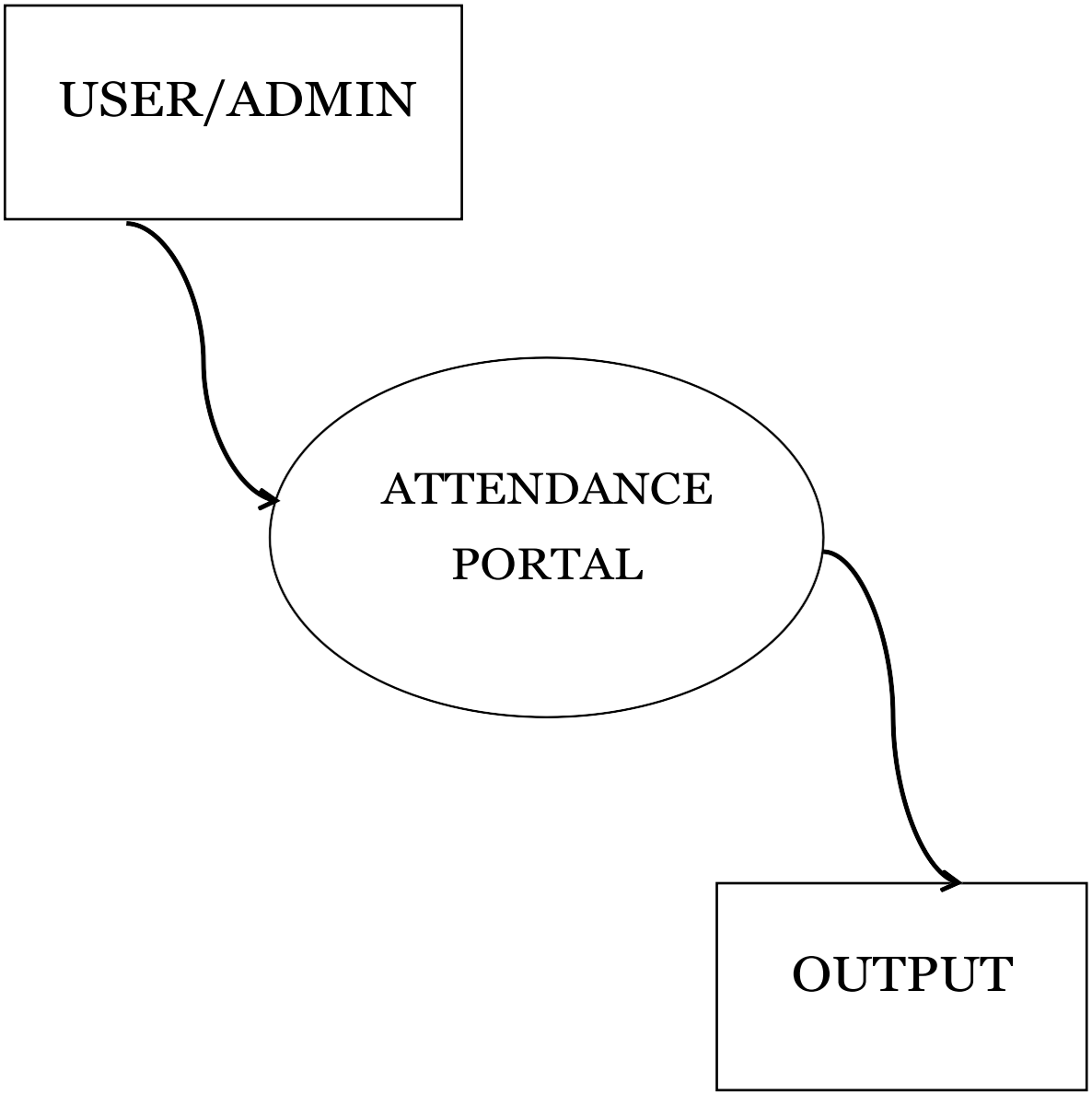
Student Attendance Management falls under the category of ONLINE ATTENDANCE MANAGEMENT SYSTEM, which proves to be handy when it comes to marking or checking the attendance. The application will eliminate the paperwork in managing student attendance record manually. This online based tool can be used by computer centers or educational institutes.

The application can run both on mobile and as a web application. As soon as a teacher or a student joins the institution, he or she is requested tosign up to the online attendance management system. It can be accessed by providing valid ID and Password of the registered user ofthe application. They can access modules such as viewing their own profile, student's profile, view attendance of students and number of students present on a particular day.

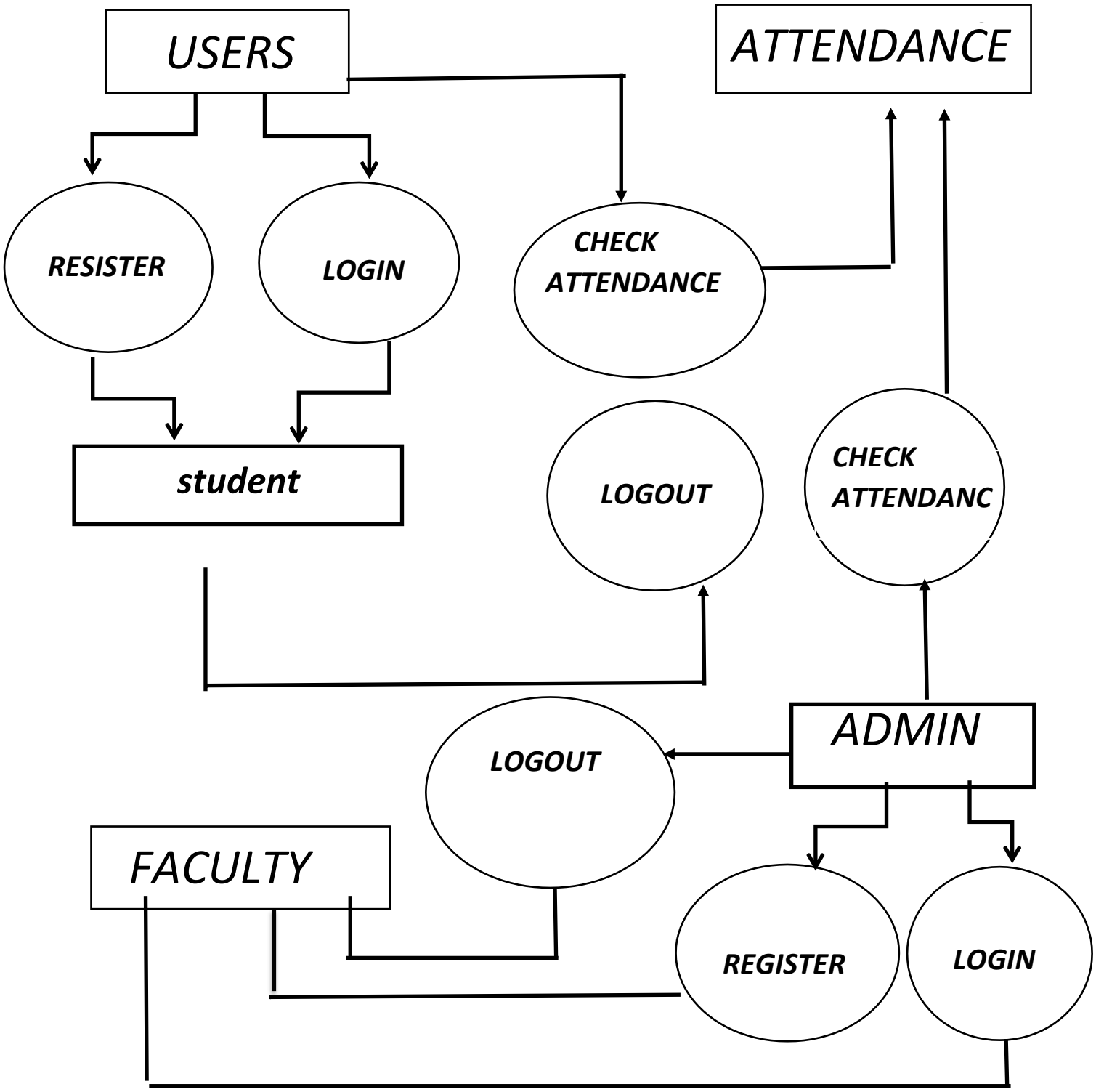
ANALYSIS

DATA FLOW DIAGRAM (DFD):

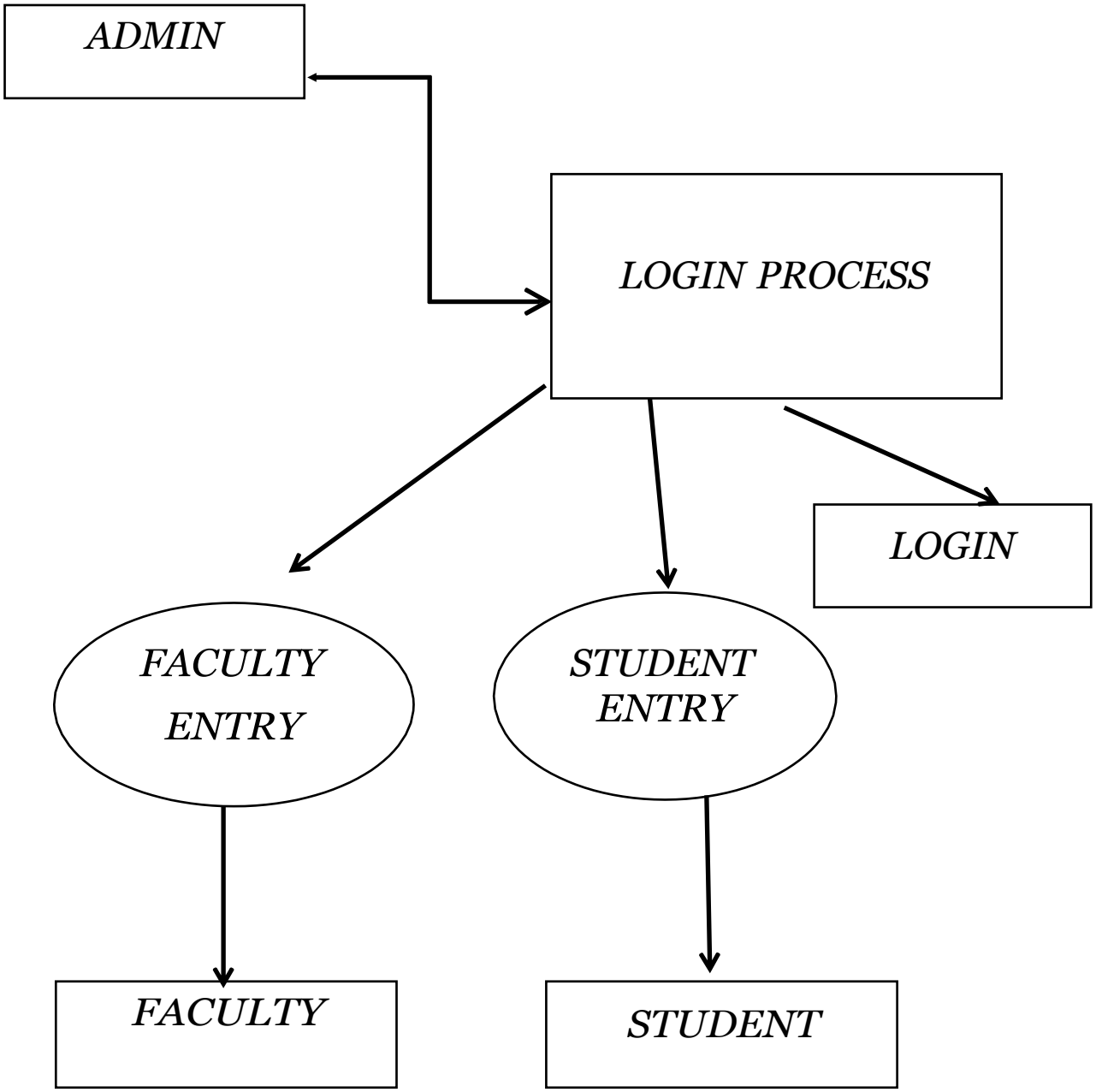
0-LEVEL DFD:



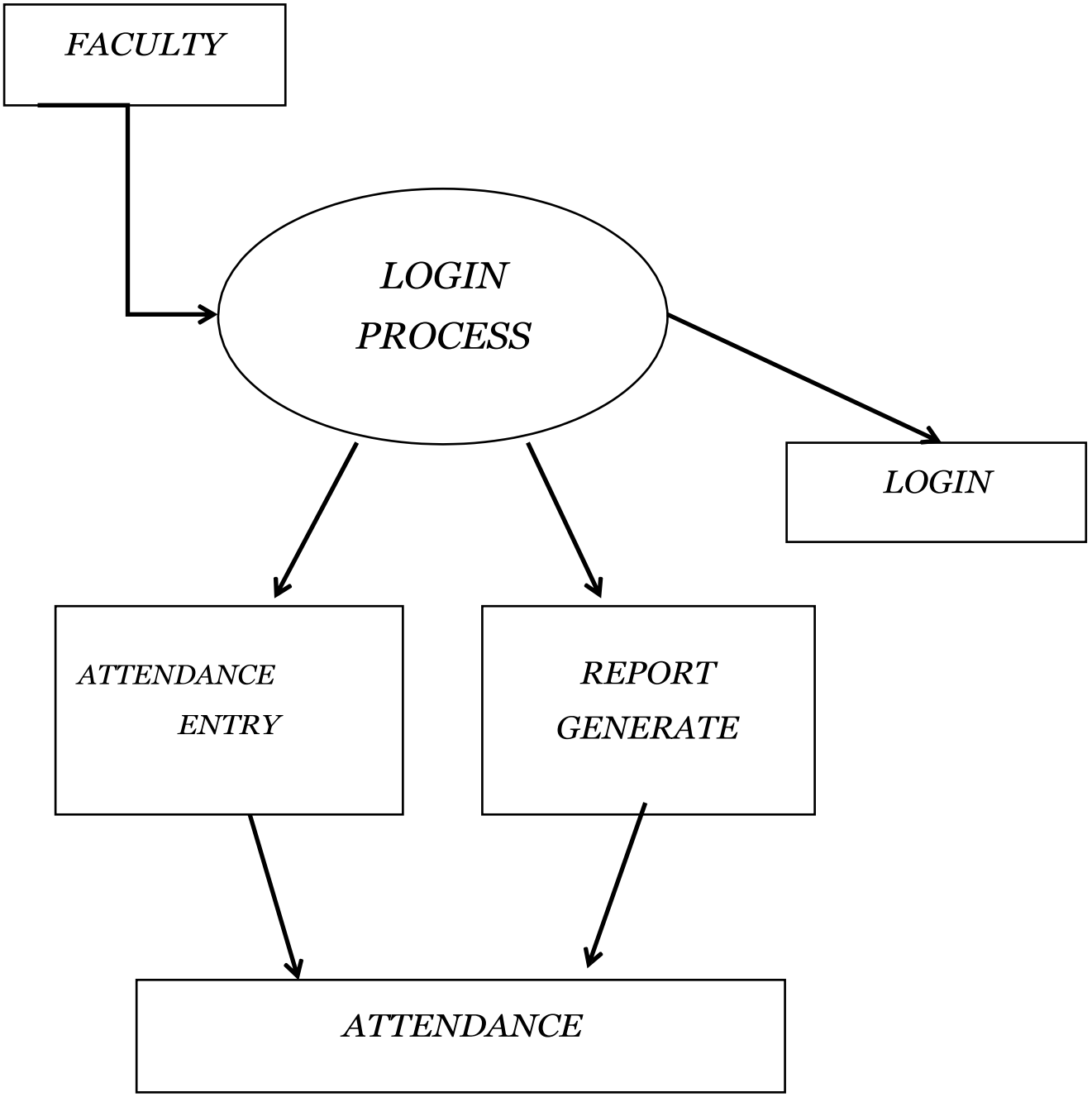
1-LEVEL DFD:


USERS
ATTENDANCE
RESISTER
LOGIN
CHECK ATTENDANCE
student
LOGOUT
CHECK ATTENDANC
ADMIN
LOGOUT
FACULTY
REGISTER
LOGIN

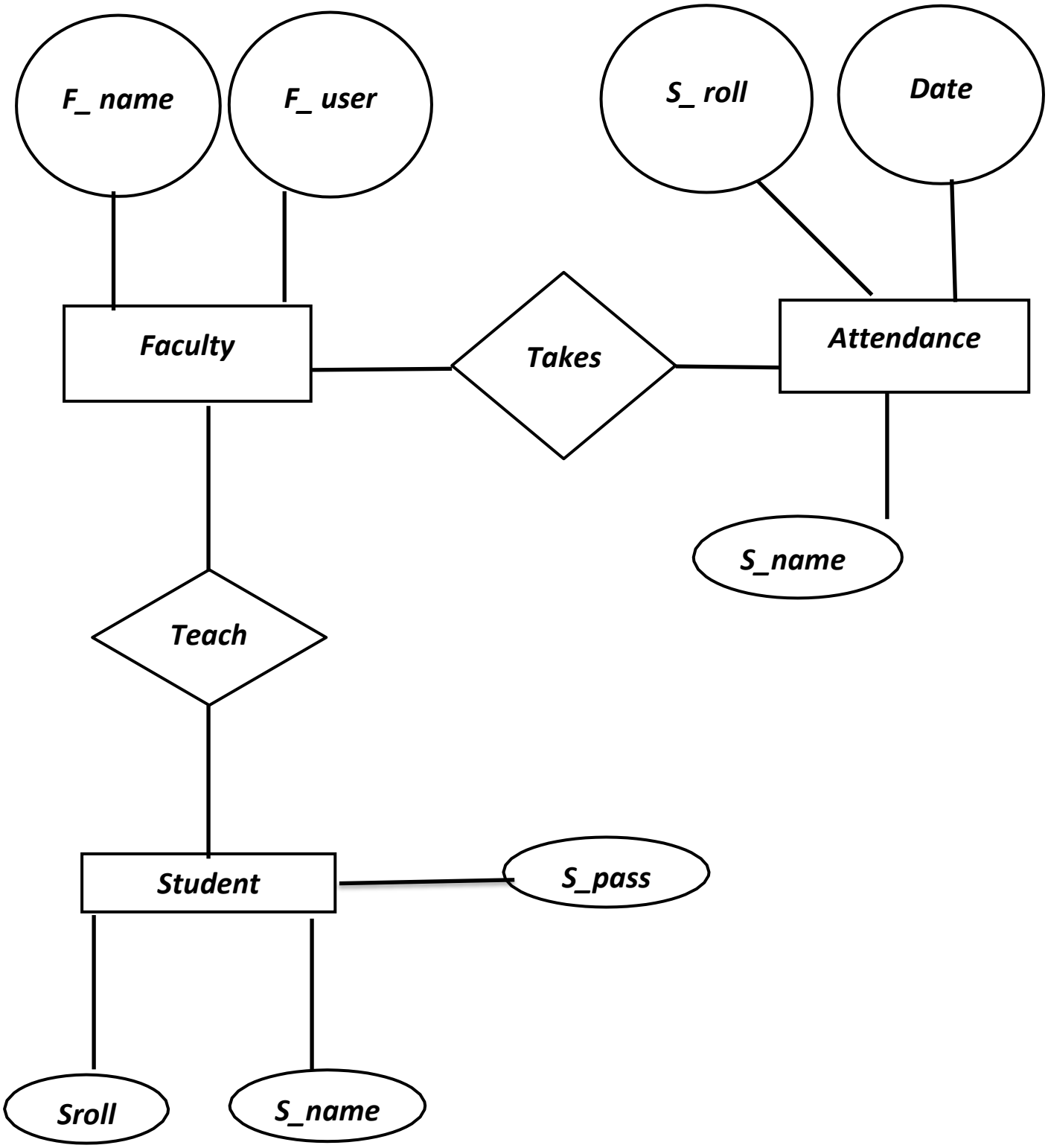
2-LEVEL DFD:


ADMIN
LOGIN PROCESS
LOGIN
FACULTY ENTRY
STUDENT ENTRY
FACULTY
STUDENT

3-LEVEL DFD:


FACULTY
LOGIN PROCESS
LOGIN
ATTENDANCE ENTRY
REPORT GENERATE
ATTENDANCE

ER-DIAGRAM:


F_ name
F_ user
S_ roll
Date
Faculty
Takes
Attendance
S_name
Teach
Student
S_pass
Sroll
S_name

Complete Structure of the Project

Module Description

The system should be designed in such a way that only authorized people should be allowed to access some modules. The records should be modified only by the administrators and no one else. The user should always be in control of the application and notvice versa.

The user interface should be consistent so that the user can handle the application with ease and speed. The application should be visual, and conceptually clear.

Administrator Module

Student Details: This module deals with the allocation of roll no. and personal details for the new batch. It generates the personal and academic details of the students.

Faculty (staff) Details: It helps to allot the subject and the subject code to the staff. It also provides the facility to have a username and password to the staff.

Timetable Details: It will retrieve the subject information from the subject information from the subject database and assign timetable to the staff. It will help the admin staff to make entry of attendance based on the subject and period allotted to the staff.

Attendance Details:

It will make the attendance database of all the students. Enter attendance to store in the database, subject, period wise into the date. It will help to get weekly reports and consolidate the attendance system.

Faculty Module

Attendance Details:

It assists the staff to mark the attendance of the students for their subject. This will authenticate the staff before making the entry.

Report Details:

Weekly reports get details of attendance from starting date to ending date and display the status. Consolidate report gets all students' attendance details from starting date to ending date status help for the eligibility criteria of the student to attend the examination.

Student Module

Generate report, get details of attendance from staring date to ending date and display the status.

Table Design of Modules

1. Faculty Table :

Field Name	Data Type
F_user	Varchar(20)
F_pass	Varchar(20)

2. Attendance Table:

Field name	Data Type
S_roll	Number
S_name	Varchar(20)
Adate	Date

3. Student Table:

Field's name	Data types
S_roll	Number
S_name	Varchar(20)
S_pass	Varchar(10)

Tools and Platform Used

HTML and CSS:

Html stands for Hyper Text Markup Language and CSS stands for Cascading Style Sheets are the crucial technologies for creating web pages. HTML supplies the structure of the page and CSS the layout, fordiversity of devices. Together with scripting and graphics, HTML and CSS are the fundamentals of building web applications and web pages.Html provides designers and developers the following facilities,

- To design forms for directing transactions with remote services, for use in making reservations, searching for information, ordering products and others
- Retrieving online information through hypertext links.
- To include video and sound clips, spreadsheets and other applications straight in their documents.
- Designers can publish online documents with text, headings,tables, photos and others.

CSS describes the web page presentation, involving layout, colors, andfonts. It enables the designer to adjust the presentation to various types of devices, like small screens, large screens or printers.

CSS is separate from HTML, and their separation makes it easy topreserve and maintain sites, share style sheets across pages, and accommodate pages to various environments.

My SQL:

SQL stands for structured query language. MySQL is an open-source relational database management system. It is a popular database for use in web applications, and is a central part of the greatly used lamp

(Linux, Apache, MySQL, Perl/PHP/Python) open – source web application software stack.
MySQL is used by many applications like Wordpress, Joomla, TYPO3, drupal, My bb, PHPbb, modx and other software. Numerous large scale websites including Google, YouTube, Facebook, Twitter and Flickr are also using MySQL.
On all platforms excluding Windows, MySQL sends with no GUI (graphical user interface) to administer MySQL databases or managing the data held within the database.
Swedish company has created MySQL which is written in C and C++.

Python:

Python can be used to build server-side web applications. While a web framework is not required to build web apps, it's rare that developers would not use existing open-source libraries to speed up their progress in getting their application working.

Python is not used in web browsers. The language executed in browsers such as chrome, Firefox and internet explorer is JavaScript. Projects such as pyjs can compile from python to JavaScript. However, most python developers write their web applications using acombination of python and JavaScript. Python is executed on the server side while JavaScript is downloaded to the client and run by the web browser.

Django:

Django is an open – source python web framework used for rapid development, pragmatic, maintainable, clean design and secures websites. A web application framework is a toolkit of all components needed for application development.
The main goal of the Django framework is to allow developers to focus on components of the application that are new instead of spending time on already developed components. Django is fully featured than many other frameworks on the market. It takes care of a

lot of hassle involved in web development, enables users to focus on developing components needed for their application.

Hardware Resource Requirements

Processor	:	1.70 GHz and Above
Main Memory	:	512 MB.
Hard Disk	:	20 GB.
Disk Space	:	100 MB.
Keyboard	:	ANY
Mouse	:	ANY
Monitor	:	ANY
Web Camera	:	ANY

Software Resource Requirements

Operating System	:	Windows family
Software	:	Web browser
Data Base	:	SQL Server2005

Scope of application

The scope of the project is the system on which the software is installed, i.e. the project is developed as a desktop application, and it will work for a particular institute. But later, the project can be modified to operate it online.

The project will have a very vast scope in future. The project can be implemented on intranet in future. Project can be updated in near future as and when requirement for the same arises, as it is very flexible in terms of expansion. With the proposed software for database space manager software ready and fully functional the client is nowable to manage and hence run the entire work in a much better, accurate and error-free manner.

The following are the future scope for the project.

- Discontinue of students eliminates potential attendance.
- Bar code reader-based attendance system.
- Individual attendance system with photo using student login.

Some future improvements can also be made to our project by makingthis biometric attendance system to make more advanced and increase its reliability and effectiveness.

A fingerprint scanner has two basic tasks which are,

1. It requires to get an image of a person finger.
2. It requires identifying and diagnosing whether the pattern of ridges and valleys in current image matches the pattern of ridges and valleys of previous scanned images.